Marcus Ziniti

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Education and Relevant Coursework:

University of Massachusetts Amherst | Commonwealth Honors College | Dan Riccio College of Engineering — Amherst, MA

Bachelor of Science in Mechanical Engineering — Expected May 2028

- GPA 3.85 | Dean's List
- UMass Aeronautics Team, UMass Rocket Team

MATLAB, Thermodynamics, Statics, Mechanics of Materials, Calc IV, Statistics, Fluid Mechanics, Dynamics

Skills:

Onshape, Solidworks, Engineering Design, Mechanical Design, MATLAB, FEA, CNC Machining, Design for Manufacturing (DFM), Basic Welding & Metalworking, Project Management, Java, C++

Engineering Experience:

Airframe Structures Lead

UMass Aeronautics Engineering Design Team | AIAA Design Build Fly — October 2024-Present

- Lead a team of 5-6 engineers responsible for the design, integration, and manufacturing of the competition airframe
- Develop aircraft configurations from aerodynamic, payload, and electrical subsystem requirements, translating design constraints into a manufacturable and flight-ready airframe
- Create and maintain aircraft CAD models in Onshape, including fuselage, tail, and wing architecture
- Coordinate with aerodynamics, payload, and electrical subteams to ensure subsystem compatibility and integration
- Lead initiatives to improve manufacturing capability through workflow optimization, standardized processes, jig and fixture integration, and a planned 3-axis cnc router
- Contribute 5-10 pages of technical content for the teams 60 page competition design report and provide design updates during technical reviews

Payload Engineer

UMass Rocket Engineering Design Team | IREC Rocket Competition — October 2025-Present

- Lead development of a MATLAB simulation to model the response of a vibration-isolation module under launch and parachute deployment conditions
- Evaluated 100 candidate spring constants using RMS vibration response metrics to quantify payload stability
- Applied a weighted value-function methodology to balance vibration reduction considerations, selecting the optimal spring configuration
- Contributed to vibration-isolation system concept development and design discussions
- Supported fabrication and assembly of the module using CNC-machined components

Projects:

Foldable Pontoon Boat Design & Fabrication

- Designed and fabricated a 9'x8' pontoon boat capable of transport in the bed of a pickup truck while retaining structural and buoyancy requirements
- Developed detailed CAD models in SolidWorks and Onshape, including structural members and flotation components
- Conducted Finite Element Analysis to evaluate structural performance under loading conditions
- Performed engineering calculations to determine buoyancy, maximum occupancy, and critical design constraints
- Executed a full engineering design cycle from concept generation through fabrication and final demonstration and presented 30-minute design reports communicating design decisions, design progress, and project outcome

Work Experience:

Project Management Intern | *Hodess Cleanrooms (Attleboro, MA)* — June 2026 - August 2026

- Lead subcontractor sourcing efforts, including identification, outreach, and qualification review for pharmaceutical and semiconductor cleanrooms projects
- Implemented BuildingConnected to source HVAC, Plumbing, and Demolition trades and procured intumescent coating contractors for a semiconductor wafer fabrication facility
- Lead a team of 4 interns in the creation of a SharePoint onboarding platform, improving knowledge transfer and training resources for future intern classes
- Developed and standardized a bid-leveling workbook under the direction of project executives, adapting industry best practices to improve subcontractor bid evaluation and procurement workflows